

How Do Different Diseases Spread, and How Can Network Science Help Us Protect Populations?

Matilde Maria
MSc Bioinformatics,
202509350

matildem.msousa@gmail.com

Lucas Caridade
MSc Bioinformatics,
202509274

lucascaridade.work@gmail.com

Rafael Correia
MSc Bioinformatics,
202207367

rafaelmdcorreia@gmail.com

ABSTRACT

abstract

1. INTRODUCTION

2. METHODOLOGY

Our question is how different types of disease spread across the transport network that connects Europe, and which network-science protection best contains them. Answering it fairly requires three components built in sequence: a transport network derived from open data (Section 2.1), a metapopulation epidemic engine that runs on it (Section 2.2), and a set of disease types and protection strategies evaluated on top (Sections 2.3 and 2.4). A single principle constrains every component: across any comparison, only the factor under study varies, while all other inputs are held fixed. The study is therefore comparative rather than predictive. We do not estimate how many people a real epidemic would infect; we quantify differences between scenarios (one strategy versus another, one disease type versus another) under identical conditions. The whole pipeline, from data retrieval through network generation to evaluation, is a single reproducible workflow (Figure 1).

2.1 Building the map

The substrate is a weighted graph $G = (V, E)$ in which nodes $v \in V$ are cities and edges $(i, j) \in E$ are travel routes carrying weight w_{ij} proportional to traffic. We construct G with NetworkX [11] from four open sources: air routes from OpenFlights [20], city populations from GeoNames, road and rail topology from OpenStreetMap and GRIP [18], and ferry routes from OpenStreetMap. Region is a parameter of construction, which is what allows us to hold the construction method fixed while varying only geography.

Because a single air layer misrepresents how an epidemic actually propagates, the same node set carries up to three layers, air, land, and water, combined into one shared-node multilayer network (Figure 2). Edge weights are layer-specific and serve as passenger-volume proxies, but they come from different recipes according to what data exists. Air weights are real route frequencies from OpenFlights, and water weights are real ferry-route frequencies counted per city pair from an OpenStreetMap ferry dump. That dump is filtered to passenger-carrying services (OSM tag

passenger not equal to no) whose endpoints are at least 5 km apart: the first filter removes freight-only crossings, since the model concerns human movement, and the distance threshold discards short river and canal crossings while retaining genuine straits and sea ferries that connect distinct cities. Land has no comparably uniform cross-region flow data, so its weights are instead modelled with a radiation kernel $K_{ij} = \text{flux}_{ij}/\text{pop}_i$ [21] computed from city populations. The constraint we impose for fairness is not that all layers share one recipe, but that each layer is built by a single method applied identically to every region (OpenFlights for air, the same global ferry dump for water, the radiation kernel for land), so that differences between regions reflect topology rather than the underlying data source; European commuting data (Eurostat) is reserved for validating the land kernel within Europe, not for parameterising any region. Node populations N_i are taken from GeoNames where available, with a degree-based proxy $N_i = P_0 + d(i) P_{\text{route}}$ as a fallback.

2.2 Running an outbreak on the map

Each city runs a compartmental model internally, while individuals move between cities along the edges of G once per day. This is a metapopulation reaction-diffusion process [8]: a local reaction updates the compartments within each city, and a mobility step couples cities through the network. Within city i the force of infection is $\lambda_i = \beta I_i/N_i$, and a daily step applies the reaction to every city and then the mobility coupling before advancing to the next day.

The mobility step is where the multilayer map earns its keep. Treating all travel as a single diffusive process is a known source of error [3], so the engine implements the two mechanisms matched to the physics of the layers, and we commit to this pair rather than collapsing them into one:

- Diffusive mobility (air, water). A fraction of each compartment leaves city i for neighbour j each day, proportional to the layer travel rate τ_ℓ and the normalised edge weight w_{ij} , and mixes into the destination population. This is the standard reaction-diffusion coupling and carries a global invasion threshold in mobility rate [9].
- Recurrent mobility (land, commuting). Commuters travel to a neighbour by day, mix there, and return home at night, so they are not relocated. We model this by coupling the force of infections rather than moving population: residents of i experience $\pi_i = \sum_j C_{ij} I_j^*/N_j^*$, where C is the per-capita com-

Table 1: Disease archetypes as compartmental model types. All five are run, from the highly transmissible immunizing type (measles-like) through latent, lethal-isolation, and waning-immunity types to the slow endemic type (cold/gonorrhea-like). Per-type rates and their literature sources are given in Section 2.5 and Appendix A.

Model	Example disease	Dynamics	R_0 (lit.)	Key feature
SIR	Measles-like	acute, immunising	12–18	remove after recovery
SIS	Common-cold-like	endemic, no immunity	1.2–1.5	return to susceptible
SEIR	COVID-like	latent then infectious	2.5–3.0	incubation compartment
SEIRS	Influenza-like	waning immunity	1.3–1.8	immunity wanes ($R \rightarrow S$)
SEIQRD	Ebola-like	latency, isolation, death	1.5–2.5	quarantine + fatality

budget we compare no intervention (control), random selection, selection by degree (the highest-traffic cities), selection by betweenness (the structural bridges), and a structural rule based on k -core and motif membership rather than raw connectivity [14]. Having identified the most effective rule, we then sweep the budget itself, from a handful of cities to a few hundred, to trace the dose–response: how many cities must be vaccinated before the outbreak is substantially contained, and where the returns begin to diminish. Coverage and per-city efficacy are held fixed throughout, so that the budget and the targeting rule are the only things that vary. The second mode is edge targeting (interdiction): we close routes rather than immunise cities, in four scenarios: (A) no intervention; (B) close all air routes while retaining land and water; (C) close all air routes within an air-only model; and (D) close only the top- k routes ranked by degree versus by betweenness. Scenarios B and C exist specifically to contrast the conclusion a single-layer model reaches with the behaviour of the full multilayer network. The two modes are evaluated as distinct levers, not stacked, so each is read in the same currency: containment per unit of a fixed budget.

2.5 Parameters and protocol

Each disease type fixes its own clinical rates from the literature (Table 1): the recovery rate γ , and where present the incubation rate σ , isolation rate κ , and waning rate ω , each the reciprocal of a measured duration. We then set the transmission rate β so that the within-city basic reproduction number matches that type’s literature R_0 . Because each city is well mixed internally, that local number is β/γ for SIR, SIS, and SEIR, and $\beta/(\gamma + \kappa)$ for the isolation type, where isolation shortens the infectious period. Worked through for the two extreme types, this gives the immunizing type (measles) $\gamma = 0.13$, $R_0 \approx 15$, hence $\beta = 1.95$; and the endemic type (gonorrhea) a long infectious duration $\gamma \approx 0.03$ with $R_0 \approx 1.4$, hence $\beta \approx 0.042$; the remaining three (COVID, Ebola, influenza) follow the same rule. We deliberately do not apply a degree-based $\langle k^2 \rangle / \langle k \rangle$ correction: that adjustment is for networks whose nodes are individuals, whereas here nodes are well-mixed cities. Every rate and R_0 traces to a specific study; Appendix A gives the measured quantity, a paraphrase of the source, and the citation for each value.

We do not calibrate to outbreak data, because the transport network carries no ground-truth epidemic to fit, so absolute incidence is illustrative rather than predictive. Every run is seeded inside the largest connected component of the substrate. Adding the sparser layers (ferries especially) brings in coastal cities that sit in small disconnected pockets, and a seed dropped in such a pocket cannot reach the mainland, so

the outbreak fizzles; mixing those fizzles with full epidemics under one average would conflate two different questions, whether an outbreak takes off and how large it grows. We therefore remove the first by always seeding where the outbreak can spread, and measure the second. A single uniform horizon of 1460 days (four years) is used for cross-disease comparability: the acute type peaks and burns out within months, while the endemic type is slow and needs years to approach its equilibrium, which a short horizon never lets it reach. Across any comparison the only inputs that change are the disease type, the protection strategy, and the layer set. We report final epidemic size, peak active infection, and peak day, and we verify the engine against analytic limits, namely single-node dynamics and the final-size correspondence between SIR and bond percolation [19].

These pieces are assembled into a single staged protocol (Figure 3), walked one factor at a time rather than as a full grid. We first run each disease with no intervention to see how it spreads; we then vaccinate, comparing the targeting rules at a fixed budget and keeping the best; we re-check that winner on the other disease type to see whether it generalises; and we trace the dose–response by sweeping the winner’s budget. Route interdiction is evaluated as the separate edge-targeting lever. Every phase is run up the realism ladder (air, then +water, then +land), so each result can be read against both the disease axis and the realism axis.

2.6 Implementation and reproducibility

The study is a single Python codebase (NumPy and NetworkX) organised as three modules, data retrieval, network generation, and epidemic evaluation, over one shared engine (the system architecture is given in Appendix B). The interface is an interactive web-based simulator that configures an outbreak and streams its progression as live epidemic curves and geographic maps, suitable for both experimentation and demonstration; the same engine and configuration system are driven programmatically to run the staged protocol over many configurations. Either way the engine, configuration, and output pipeline are identical, so results do not depend on how a simulation is launched, and every run is fully determined by its configuration and random seed.

3. DISCUSSION

4. CONCLUSION

5. REFERENCES

- [1] Y. Alimohamadi, M. Taghdir, and M. Sepandi. Estimate of the basic reproduction number for covid-19: a

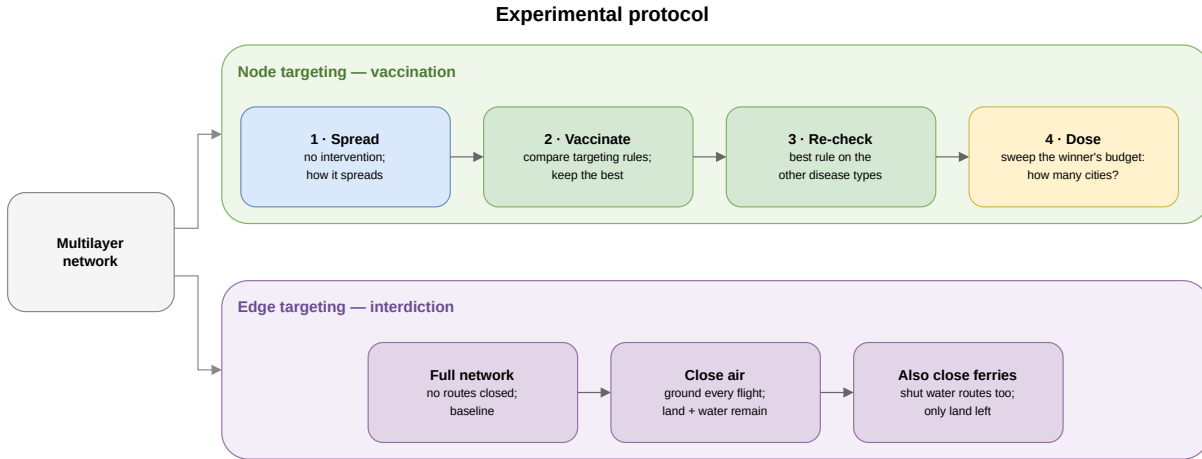


Figure 3: Experimental protocol. From one multilayer network the study splits into two protection levers. Node targeting (vaccination) is walked one factor at a time: spread, then vaccinate (keep the best targeting rule), then re-check that winner on the other diseases, then sweep its budget (dose). Edge targeting (interdiction) progressively closes routes (air, then ferries). Both levers are run for all five disease types, each up the realism ladder air $\rightarrow$ +water $\rightarrow$ +land.

- systematic review and meta-analysis. *Journal of Preventive Medicine and Public Health*, 53(3):151–157, 2020.
- [2] C. L. Althaus. Estimating the reproduction number of ebola virus (ebov) during the 2014 outbreak in west africa. *PLoS Currents Outbreaks*, 2014.
 - [3] D. Balcan and A. Vespignani. Phase transitions in contagion processes mediated by recurrent mobility patterns. *Nature Physics*, 7(7):581–586, 2011.
 - [4] L. A. Barbee, O. O. Soge, C. M. Khosropour, M. Haglund, W. Yeung, J. Hughes, and M. R. Golden. The duration of pharyngeal gonorrhea: a natural history study. *Clinical Infectious Diseases*, 73(4):575–582, 2021.
 - [5] M. Biggerstaff, S. Cauchemez, C. Reed, M. Gambhir, and L. Finelli. Estimates of the reproduction number for seasonal, pandemic, and zoonotic influenza: a systematic review of the literature. *BMC Infectious Diseases*, 14:480, 2014.
 - [6] F. Carrat, E. Vergu, N. M. Ferguson, M. Lemaitre, S. Cauchemez, S. Leach, and A.-J. Valleron. Time lines of infection and disease in human influenza: a review of volunteer challenge studies. *American Journal of Epidemiology*, 167(7):775–785, 2008.
 - [7] Centers for Disease Control and Prevention. Clinical overview of measles. <https://www.cdc.gov/measles/hcp/clinical-overview/index.html>. Accessed 2026.
 - [8] V. Colizza, R. Pastor-Satorras, and A. Vespignani. Reaction–diffusion processes and metapopulation models in heterogeneous networks. *Nature Physics*, 3(4):276–282, 2007.
 - [9] V. Colizza and A. Vespignani. Invasion threshold in heterogeneous metapopulation networks. *Physical Review Letters*, 99(14):148701, 2007.
 - [10] F. M. Guerra, S. Bolotin, G. Lim, J. Heffernan, S. L. Deeks, Y. Li, and N. S. Crowcroft. The basic reproduction number (r_0) of measles: a systematic review. *The Lancet Infectious Diseases*, 17(12):e420–e428, 2017.
 - [11] A. A. Hagberg, D. A. Schult, and P. J. Swart. Exploring network structure, dynamics, and function using NetworkX. In G. Varoquaux, T. Vaught, and J. Millman, editors, *Proceedings of the 7th Python in Science Conference (SciPy)*, pages 11–15, Pasadena, CA, USA, 2008.
 - [12] X. He, E. H. Y. Lau, P. Wu, X. Deng, J. Wang, X. Hao, Y. C. Lau, J. Y. Wong, Y. Guan, X. Tan, X. Mo, et al. Temporal dynamics in viral shedding and transmissibility of covid-19. *Nature Medicine*, 26(5):672–675, 2020.
 - [13] H. W. Hethcote and J. A. Yorke. *Gonorrhea Transmission Dynamics and Control*, volume 56 of *Lecture Notes in Biomathematics*. Springer-Verlag, 1984.
 - [14] M. Kitsak, L. K. Gallos, S. Havlin, F. Liljeros, L. Muchnik, H. E. Stanley, and H. A. Makse. Identification of influential spreaders in complex networks. *Nature Physics*, 6(11):888–893, 2010.
 - [15] S. A. Lauer, K. H. Grantz, Q. Bi, F. K. Jones, Q. Zheng, H. R. Meredith, A. S. Azman, N. G. Reich, and J. Lessler. The incubation period of coronavirus disease 2019 (covid-19) from publicly reported confirmed cases: estimation and application. *Annals of Internal Medicine*, 172(9):577–582, 2020.
 - [16] J. Legrand, R. F. Grais, P.-Y. Boelle, A.-J. Valleron, and A. Flahault. Understanding the dynamics of ebola epidemics. *Epidemiology & Infection*, 135(4):610–621, 2007.
 - [17] J. Lessler, N. G. Reich, R. Brookmeyer, T. M. Perl, K. E. Nelson, and D. A. T. Cummings. Incubation periods of acute respiratory viral infections: a systematic review. *The Lancet Infectious Diseases*, 9(5):291–300, 2009.

- [18] J. R. Meijer, M. A. J. Huijbregts, K. C. G. J. Schotten, and A. M. Schipper. Global patterns of current and future road infrastructure. *Environmental Research Letters*, 13(6):064006, 2018.
- [19] M. E. J. Newman. Spread of epidemic disease on networks. *Physical Review E*, 66(1):016128, 2002.
- [20] OpenFlights. OpenFlights: Airport, airline and route data. <https://openflights.org/data.html>, 2024. Accessed June 2026.
- [21] F. Simini, M. C. González, A. Maritan, and A.-L. Barabási. A universal model for mobility and migration patterns. *Nature*, 484(7392):96–100, 2012.
- [22] WHO Ebola Response Team. Ebola virus disease in west africa—the first 9 months of the epidemic and forward projections. *New England Journal of Medicine*, 371(16):1481–1495, 2014.
- [23] R. G. Woolthuis, J. Wallinga, and M. van Boven. Variation in loss of immunity shapes influenza epidemics and the impact of vaccination. *BMC Infectious Diseases*, 17:632, 2017.

APPENDIX

A. DISEASE PARAMETER SOURCES

Every parameter value is taken from the literature, not tuned. Each clinical rate is the reciprocal of a measured duration of the exemplar disease (rate = 1/duration); each basic reproduction number R_0 comes from a separate epidemiological study; and the transmission rate β is derived from both, $\beta = R_0\gamma$ (or $\beta = R_0(\gamma + \kappa)$ for the isolation type). Table 2 gives the source for every value. Notably, measles and Ebola have similar infectious periods (~ 8 – 10 d) yet R_0 differs roughly sevenfold (≈ 15 vs ≈ 2): the infectious period does not set R_0 ; together with β it produces it.

B. IMPLEMENTATION ARCHITECTURE

The interactive simulator and the programmatic staged-protocol runs share one engine, configuration system, and results pipeline (Figure 4), so the demonstration and the study cannot drift apart. (This figure is a working schematic and will be refined.)

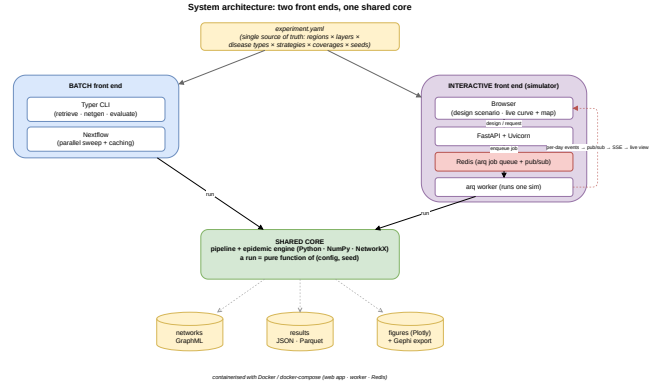


Figure 4: System architecture: one shared engine behind the interactive simulator and the programmatic runs, writing to a common results pipeline.

Table 2: Source for every disease-type parameter. Rates are per day. All five types are run, spanning the immunizing (measles) to the endemic (gonorrhea) ends of the spectrum.

Disease (model)	Value	Measured quantity	Source (paraphrased)
Measles (SIR)	$\gamma = 0.13$ $R_0 = 12\text{--}18$ $\omega = 0$	infectious ~ 8 d R_0 immunity	contagious 4 d before to 4 d after rash [7] “often cited to be 12–18” (review of 58 estimates) [10] natural infection confers life-long immunity [7]
COVID (SEIR)	$\sigma = 0.20$ $\gamma = 0.13$ $R_0 = 2\text{--}3.3$	incubation 5.1 d infectious ~ 1 wk R_0	“mean incubation period 5.1 days (95% CI 4.5–5.8)” [15] infectiousness peaks near onset, low by ~ 7 d [12] “pooled R_0 3.32 (95% CI 2.81–3.82)” [1]
Ebola (SEIQRD)	$\sigma = 0.09$ γ, κ struct. $\mu = 0.71$ $R_0 = 1.5\text{--}2.5$	incubation 11.4 d onset $\rightarrow$ hospital ~ 5 d case-fatality R_0 (SEIR fit)	“incubation period was 11.4 days” [22] SEIHFR; community / hospital / burial transmission [16] “case fatality rate 70.8% (95% CI 69–73)” [22] 1.51 Guinea, 2.53 Sierra Leone, 1.59 Liberia [2]
Gonorrhea (SIS)	SIS, $\omega = \infty$ $\gamma \approx 0.03$	immunity, latency untreated duration	“negligible protective immunity, negligible latent period” [13] median untreated pharyngeal gonorrhea ≈ 16 weeks [4]
Influenza (SEIRS)	$\sigma = 0.70$ $\gamma = 0.21$ $\omega \approx 0.001$ $R_0 = 1.3\text{--}1.8$	incubation 1.4 d shedding 4.8 d immunity R_0 (review)	“median incubation period for influenza A 1.4 days” [17] “viral shedding averaged 4.80 days (95% CI 4.31–5.29)” [6] influenza A immunity wanes in 2–7 yr via antigenic drift [23] median $R \approx 1.80$ for the 1918 pandemic (IQR 1.47–2.27) [5]